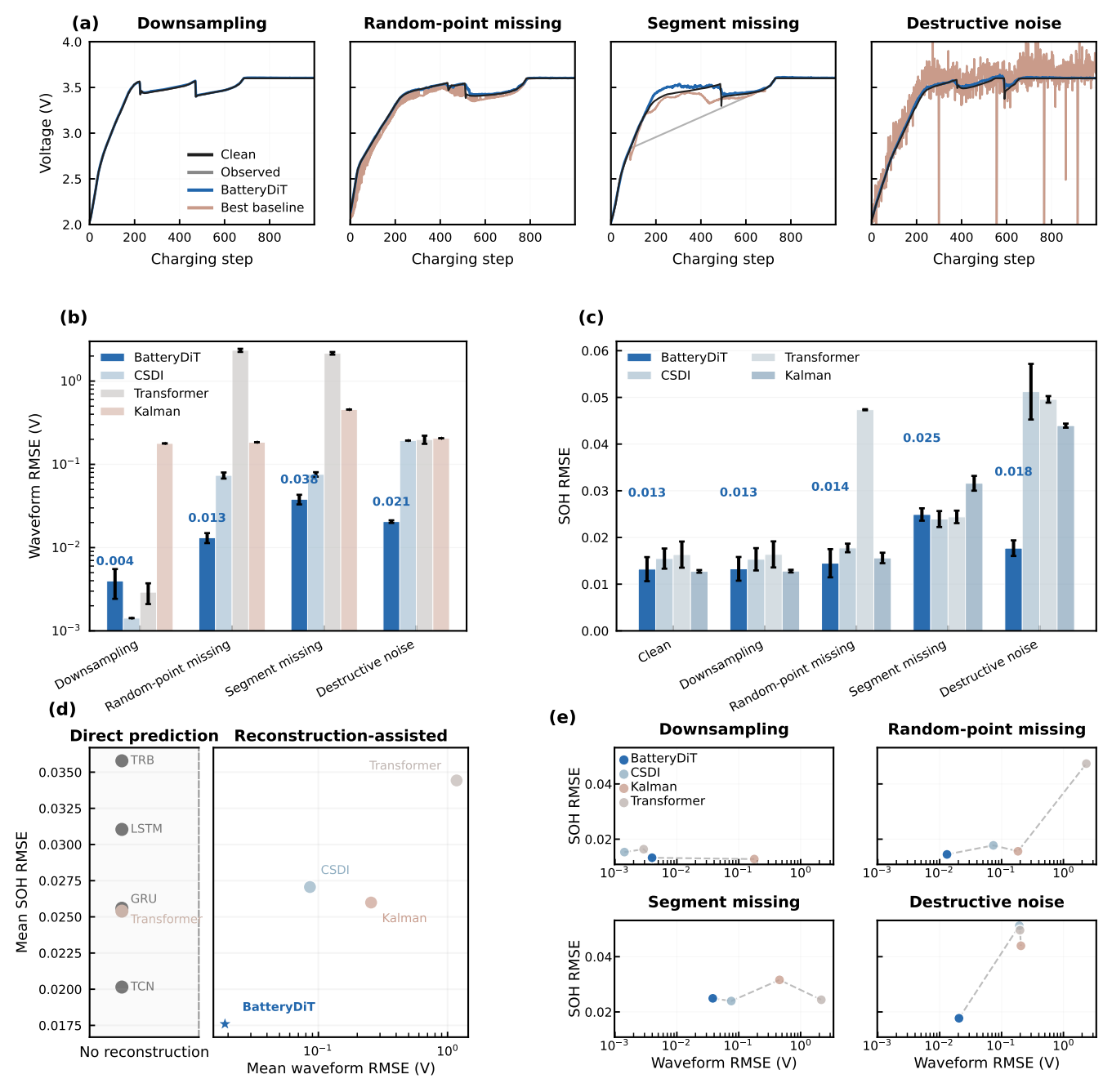

(a)
Downsampling
Random-point missing
Segment missing
Destructive noise
Voltage (V)
Charging step
Clean
Observed
BatteryDiT
Best baseline
(b)
Waveform RMSE (V)
BatteryDiT
CSDI
Transformer
Kalman
0.004
0.013
0.038
0.021
Downsampling
Random-point missing
Segment missing
Destructive noise
(c)
SOH RMSE
0.013
0.013
0.014
0.025
0.018
Clean
Downsampling
Random-point missing
Segment missing
Destructive noise
(d)
Direct prediction
Reconstruction-assisted
Mean SOH RMSE
TRB
LSTM
GRU
Transformer
TCN
CSDI
Kalman
BatteryDiT
No reconstruction
Mean waveform RMSE (V)
(e)
Downsampling
Random-point missing
Segment missing
Destructive noise
SOH RMSE
Waveform RMSE (V)